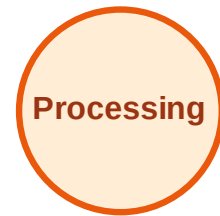


BFS Traversal

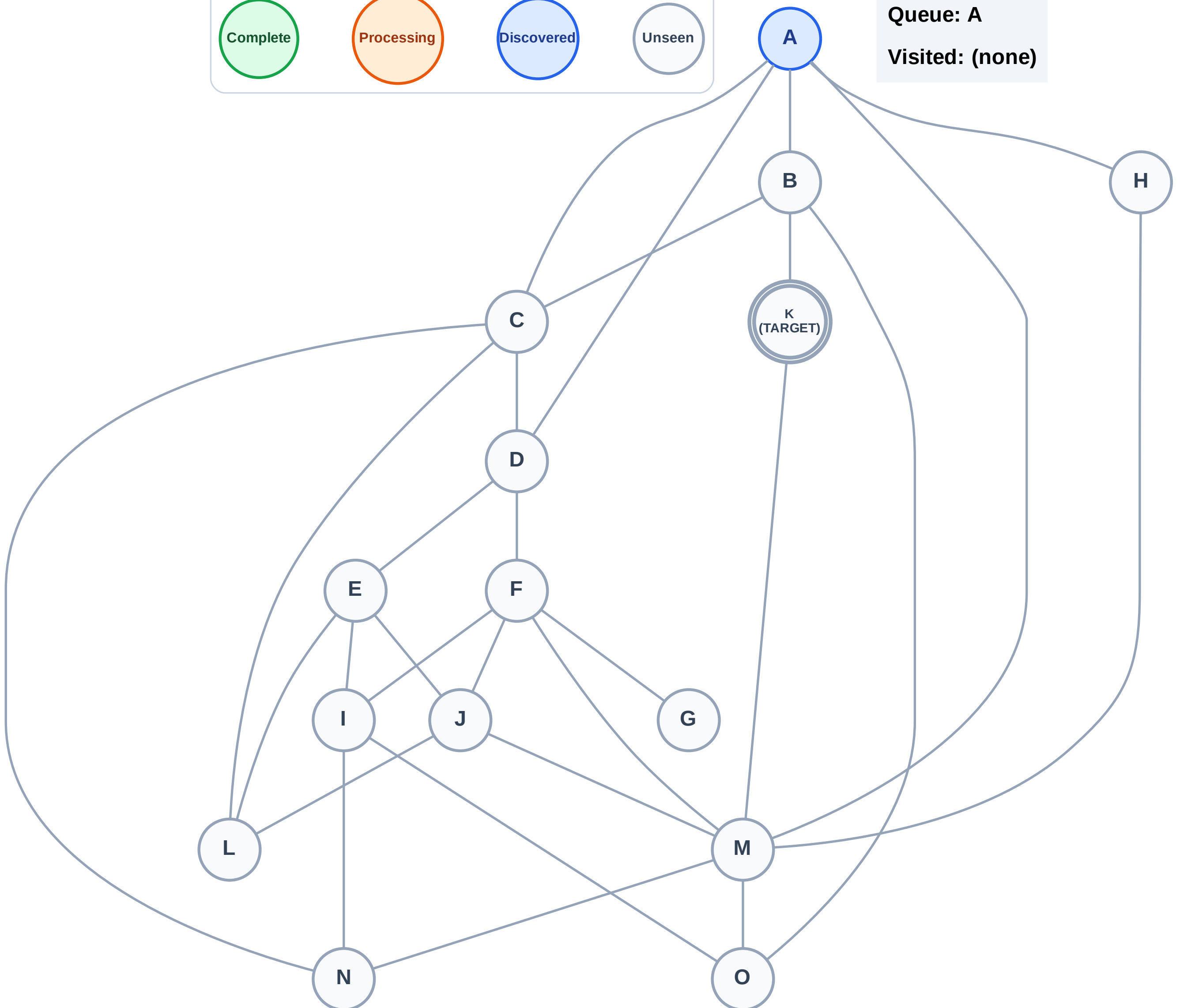
Step 1: Start at A (Seeking target node K)

Legend



Queue: A

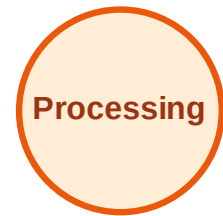
Visited: (none)



BFS Traversal

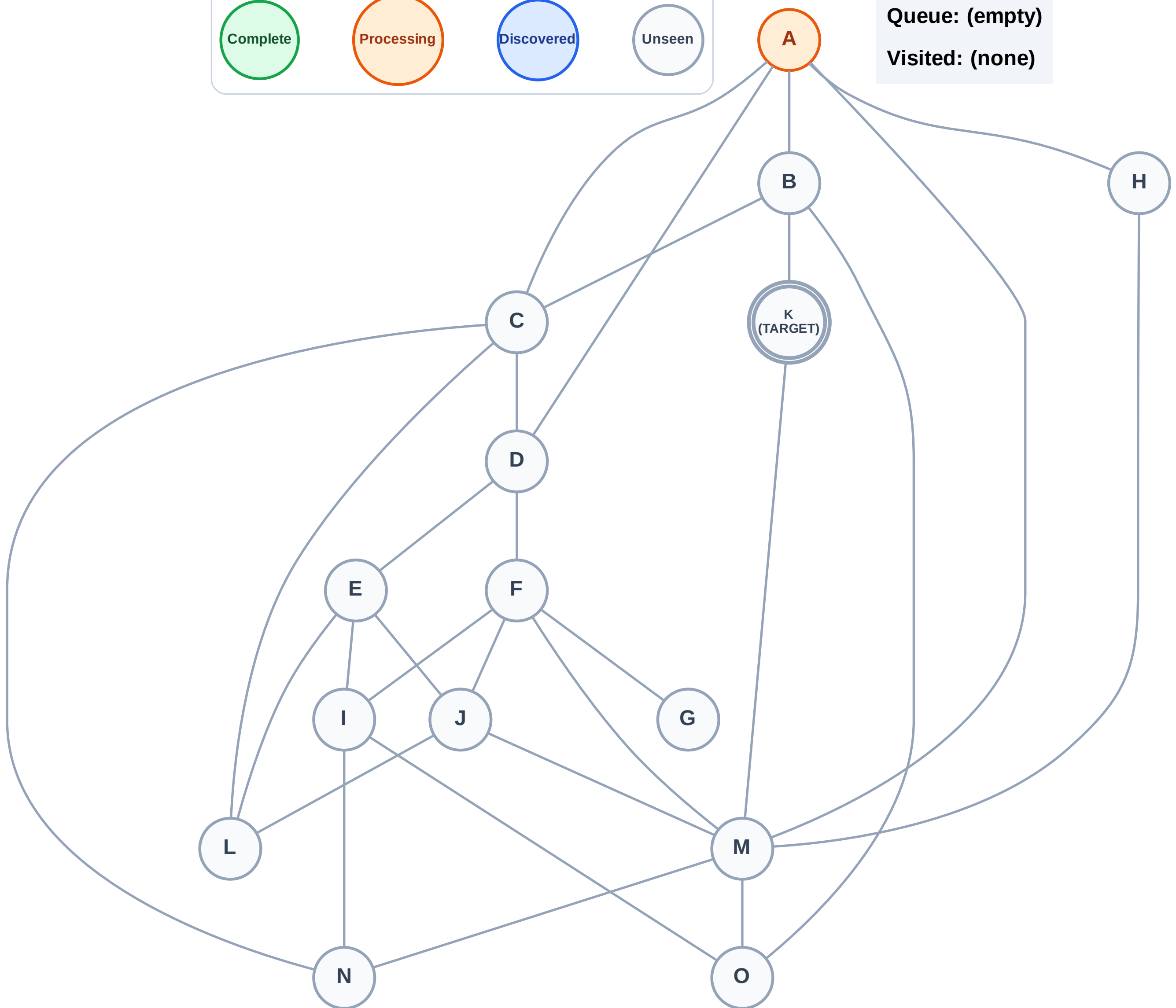
Step 2: Process node A

Legend



Queue: (empty)

Visited: (none)



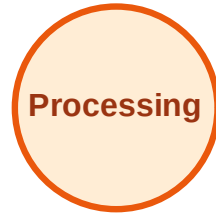
BFS Traversal

Step 3: Discover B, C, D, H, M from A

Legend



Complete



Processing



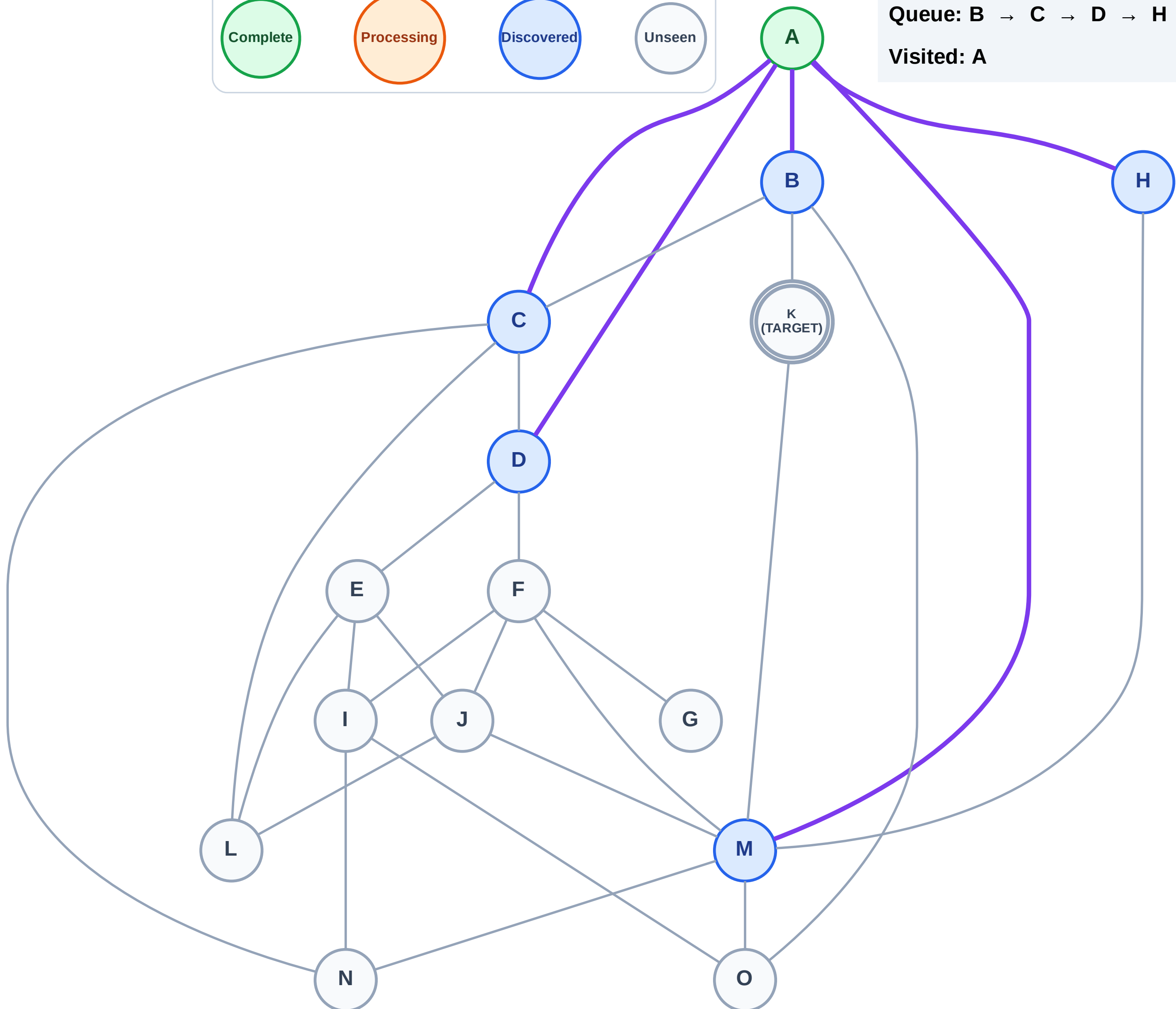
Discovered



Unseen

Queue: B → C → D → H → M

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

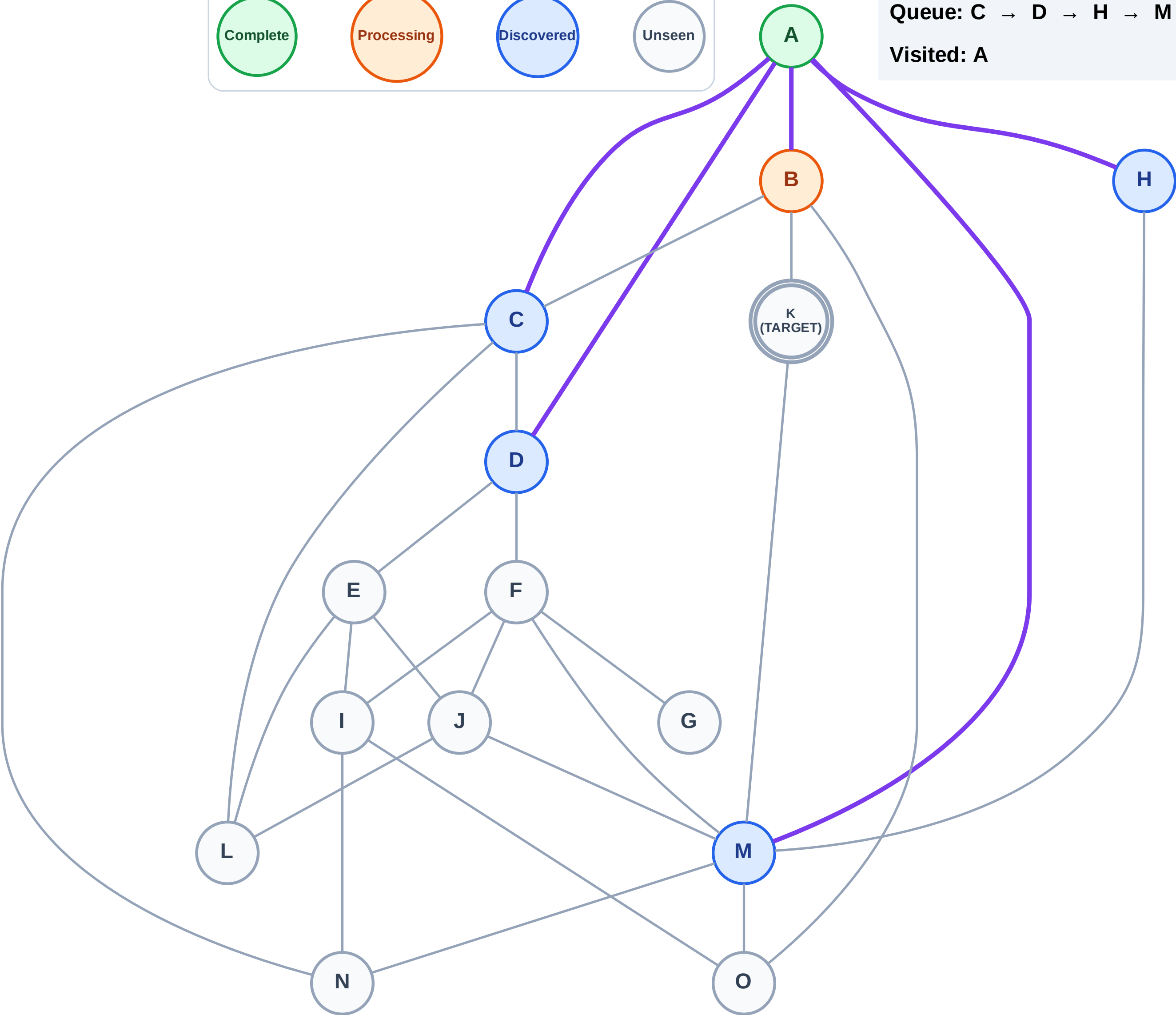
Processing

Discovered

Unseen

Queue: C → D → H → M

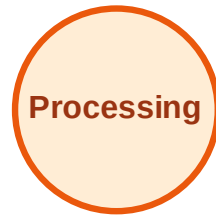
Visited: A



BFS Traversal

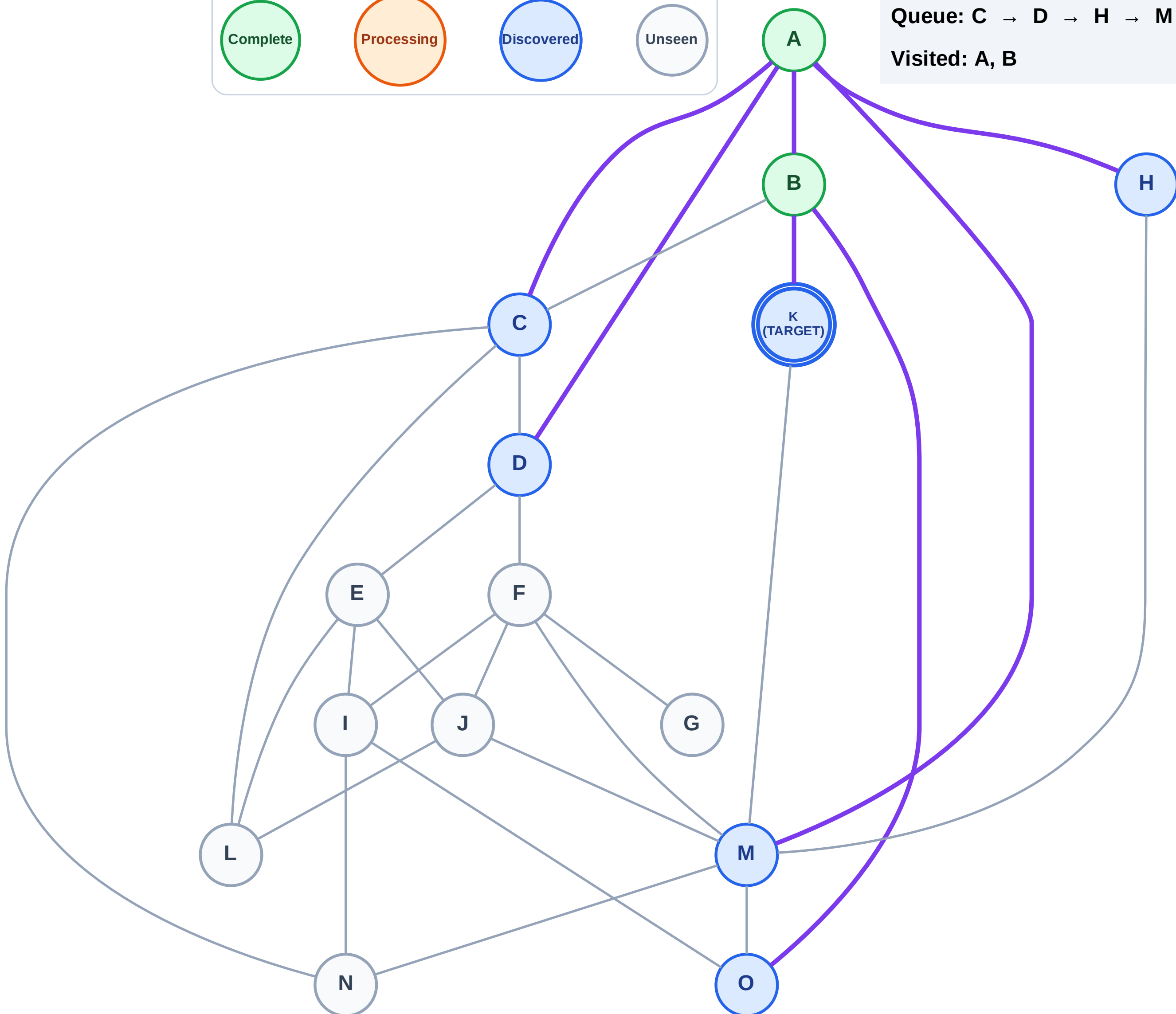
Step 5: Discover K, O from B

Legend



Queue: C → D → H → M → K → O

Visited: A, B



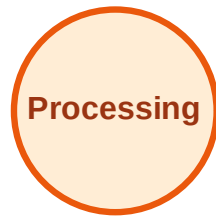
BFS Traversal

Step 6: Process node C

Legend



Complete



Processing



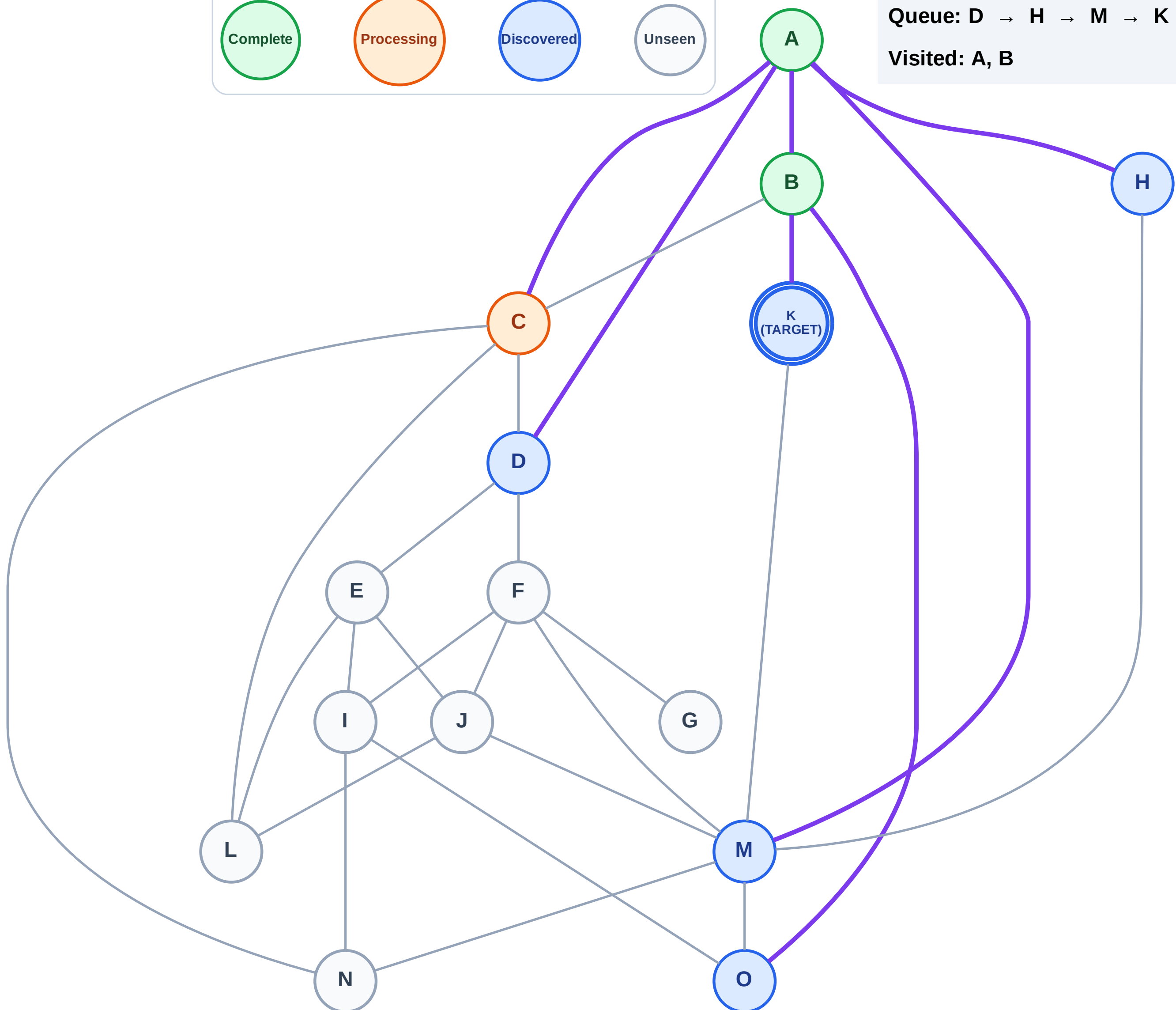
Discovered



Unseen

Queue: D → H → M → K → O

Visited: A, B



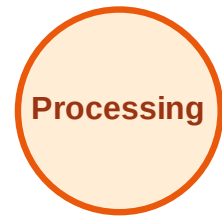
BFS Traversal

Step 7: Discover L, N from C

Legend



Complete



Processing



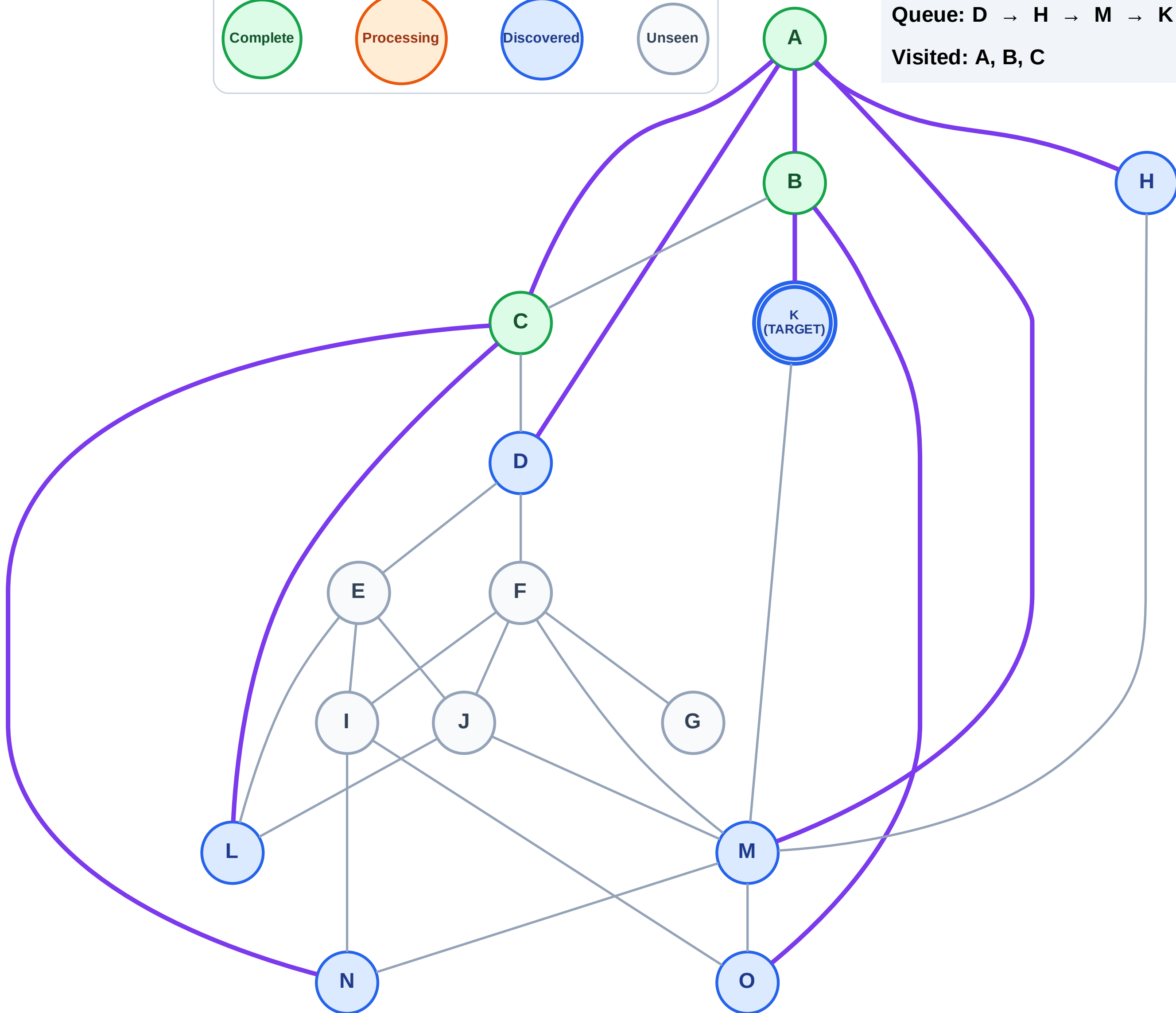
Discovered



Unseen

Queue: D → H → M → K → O → L → N

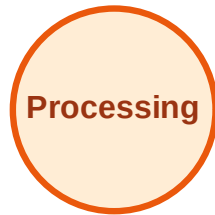
Visited: A, B, C



BFS Traversal

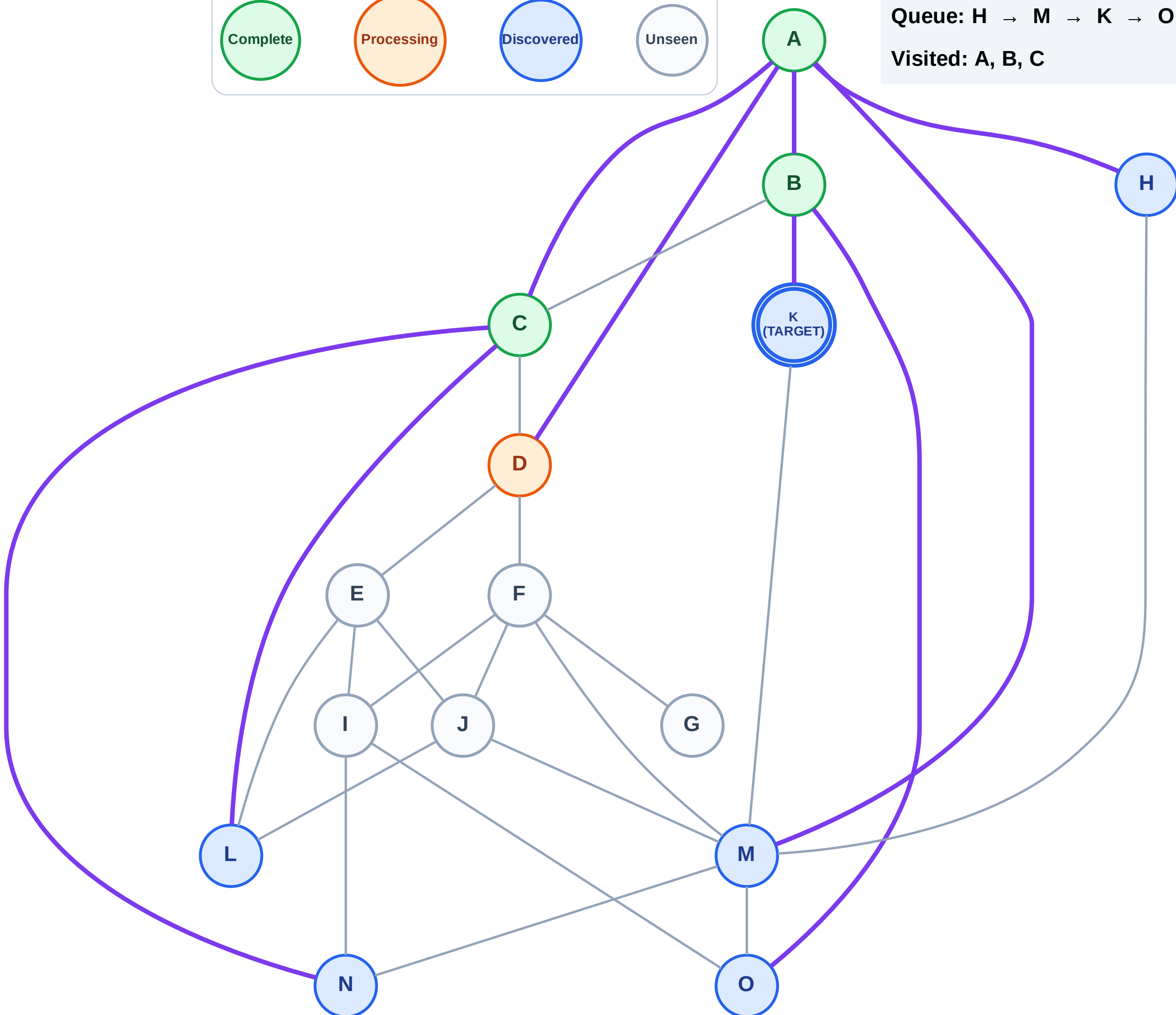
Step 8: Process node D

Legend



Queue: H → M → K → O → L → N

Visited: A, B, C



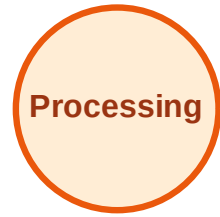
BFS Traversal

Step 9: Discover E, F from D

Legend



Complete



Processing



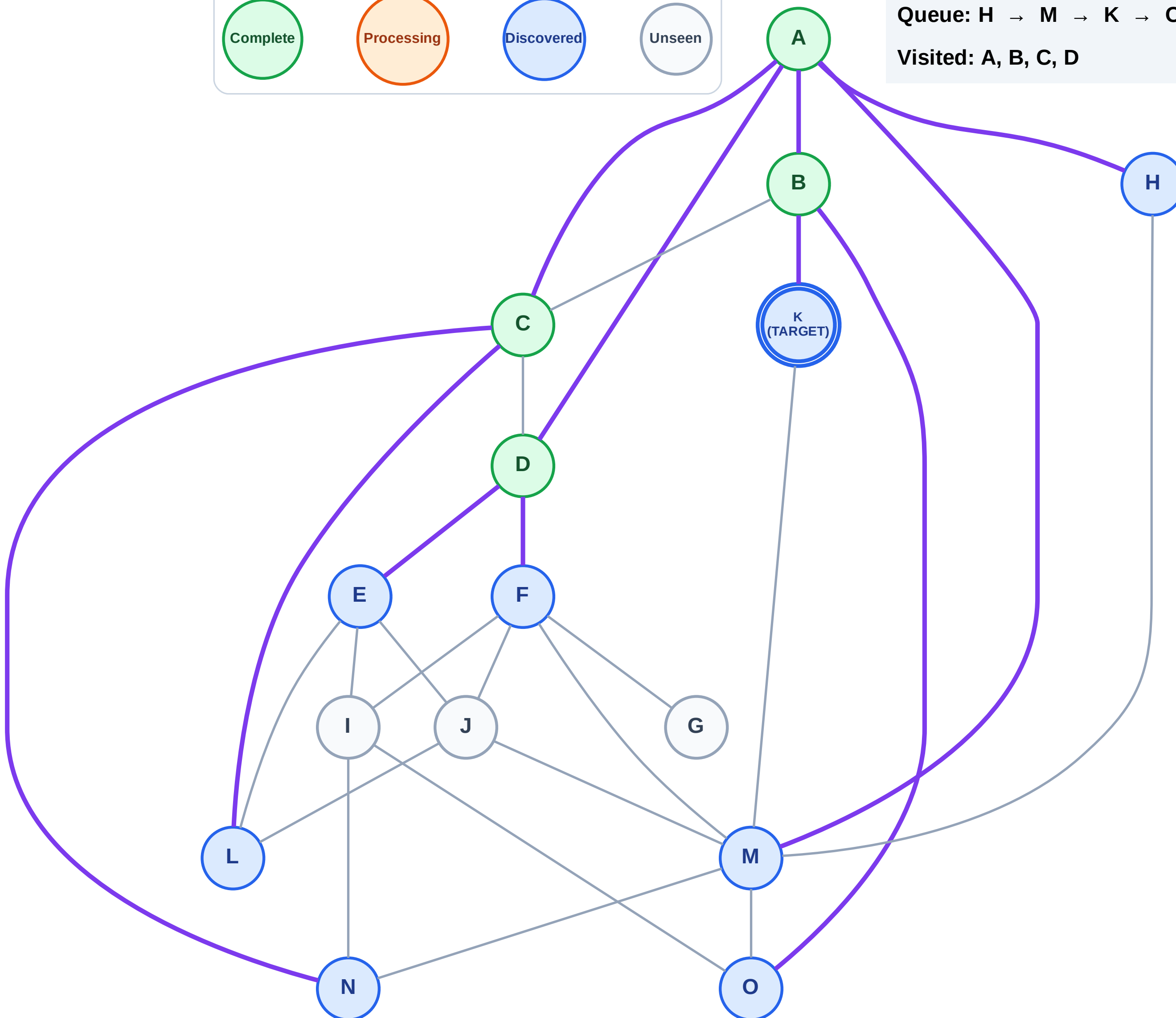
Discovered



Unseen

Queue: H → M → K → O → L → N → E → F

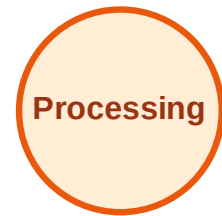
Visited: A, B, C, D



BFS Traversal

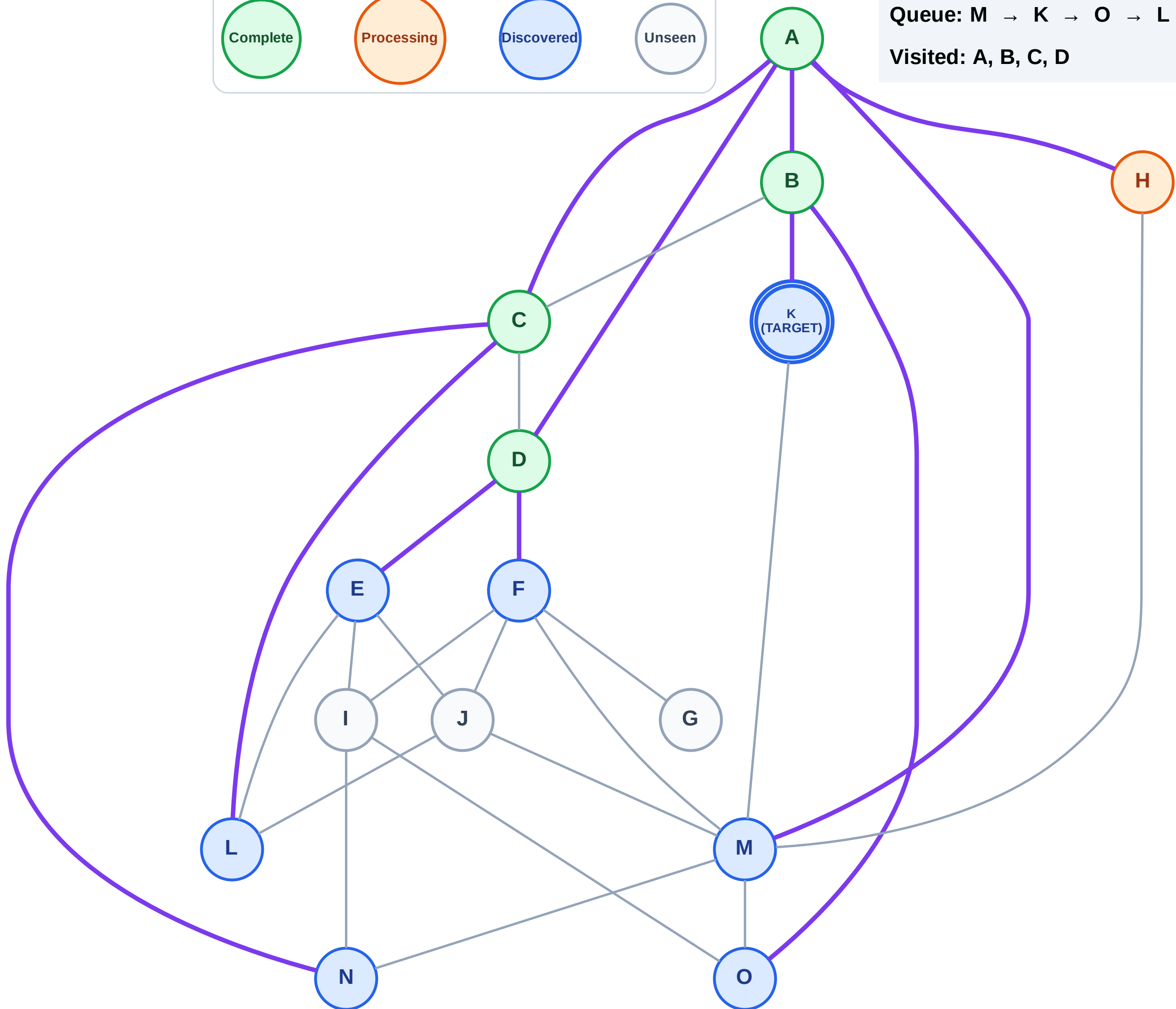
Step 10: Process node H

Legend



Queue: M → K → O → L → N → E → F

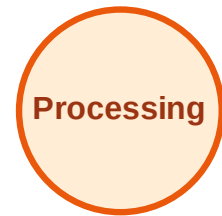
Visited: A, B, C, D



BFS Traversal

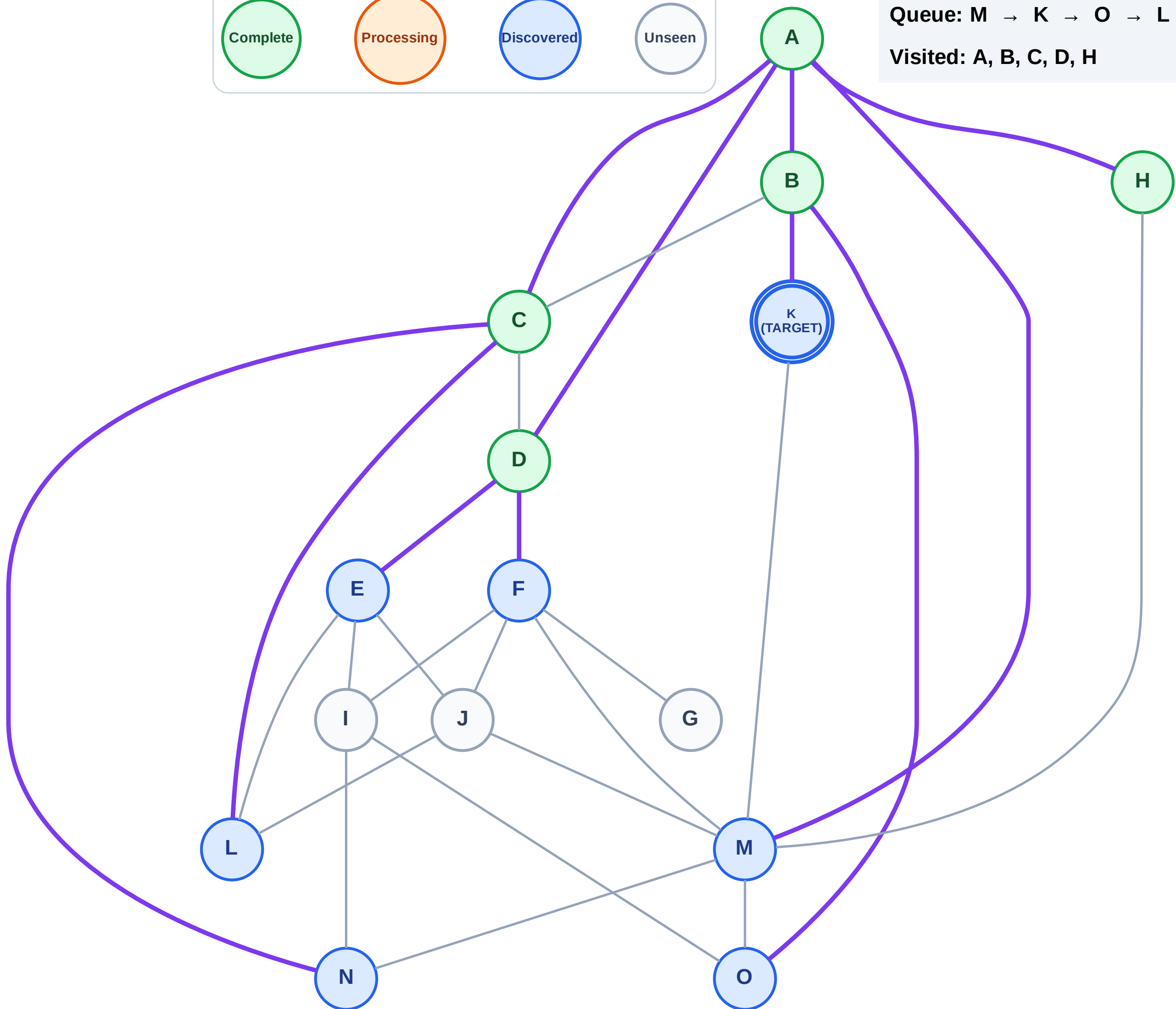
Step 11: No new neighbors from H

Legend



Queue: M → K → O → L → N → E → F

Visited: A, B, C, D, H



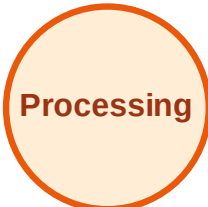
BFS Traversal

Step 12: Process node M

Legend



Complete



Processing



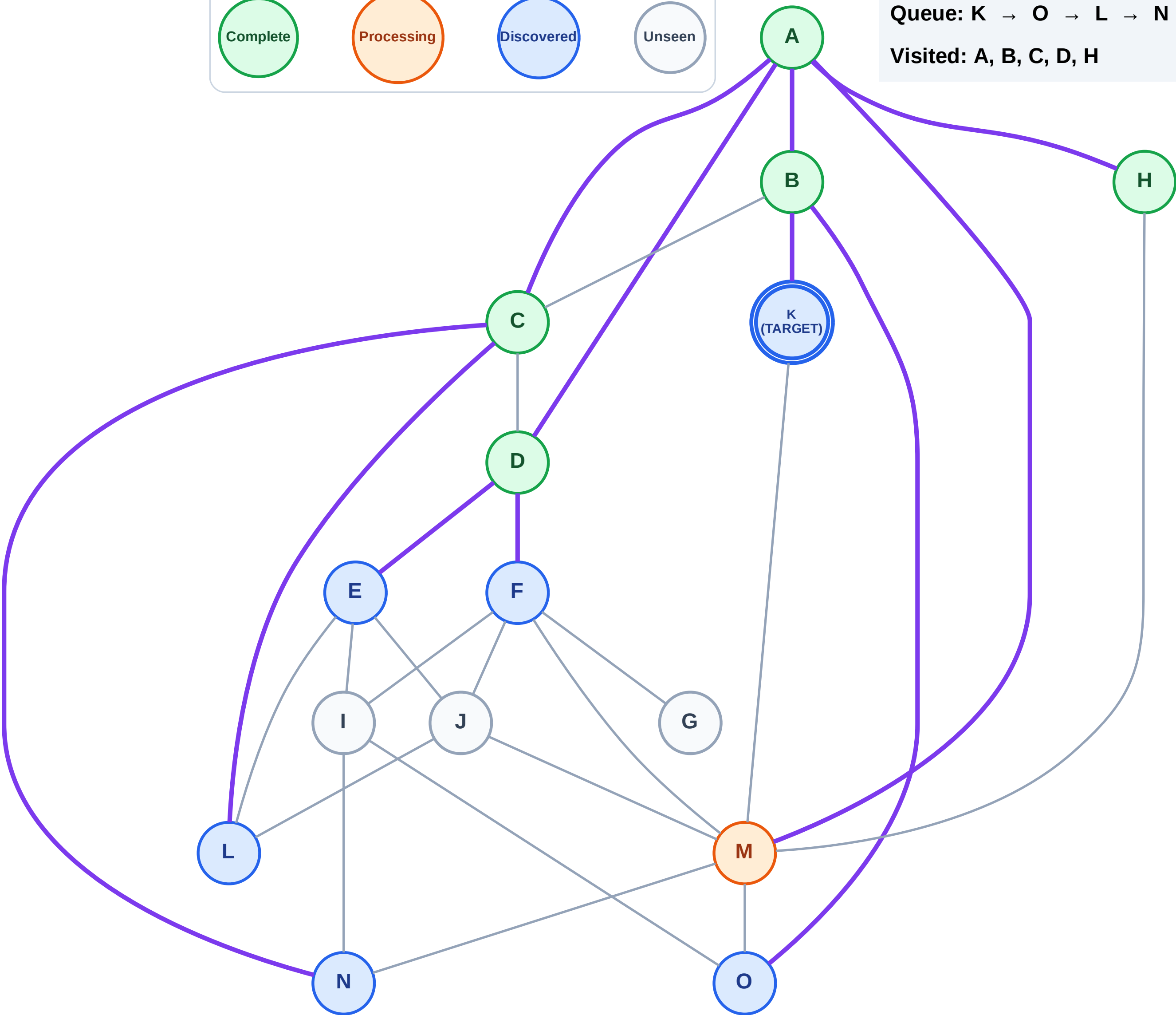
Discovered



Unseen

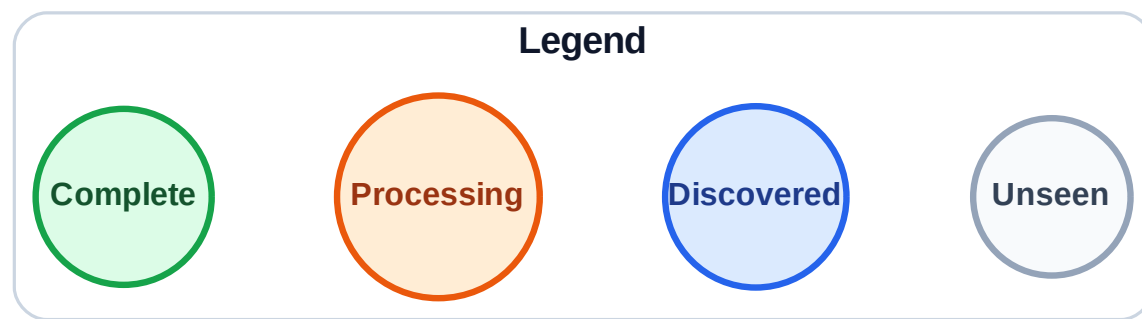
Queue: K → O → L → N → E → F

Visited: A, B, C, D, H



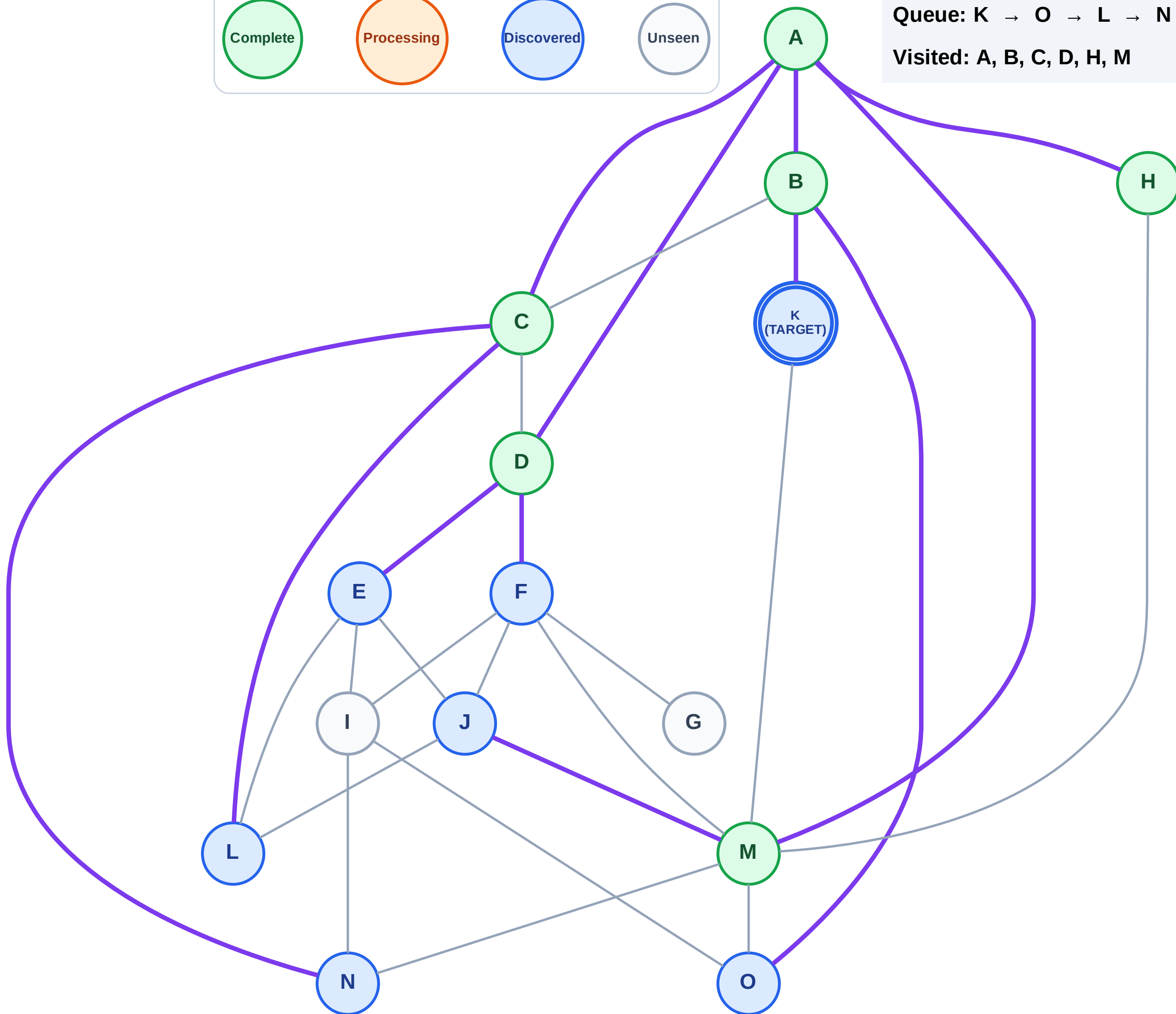
BFS Traversal

Step 13: Discover J from M



Queue: K → O → L → N → E → F → J

Visited: A, B, C, D, H, M



BFS Traversal

Step 14: Process node K

Legend

Complete

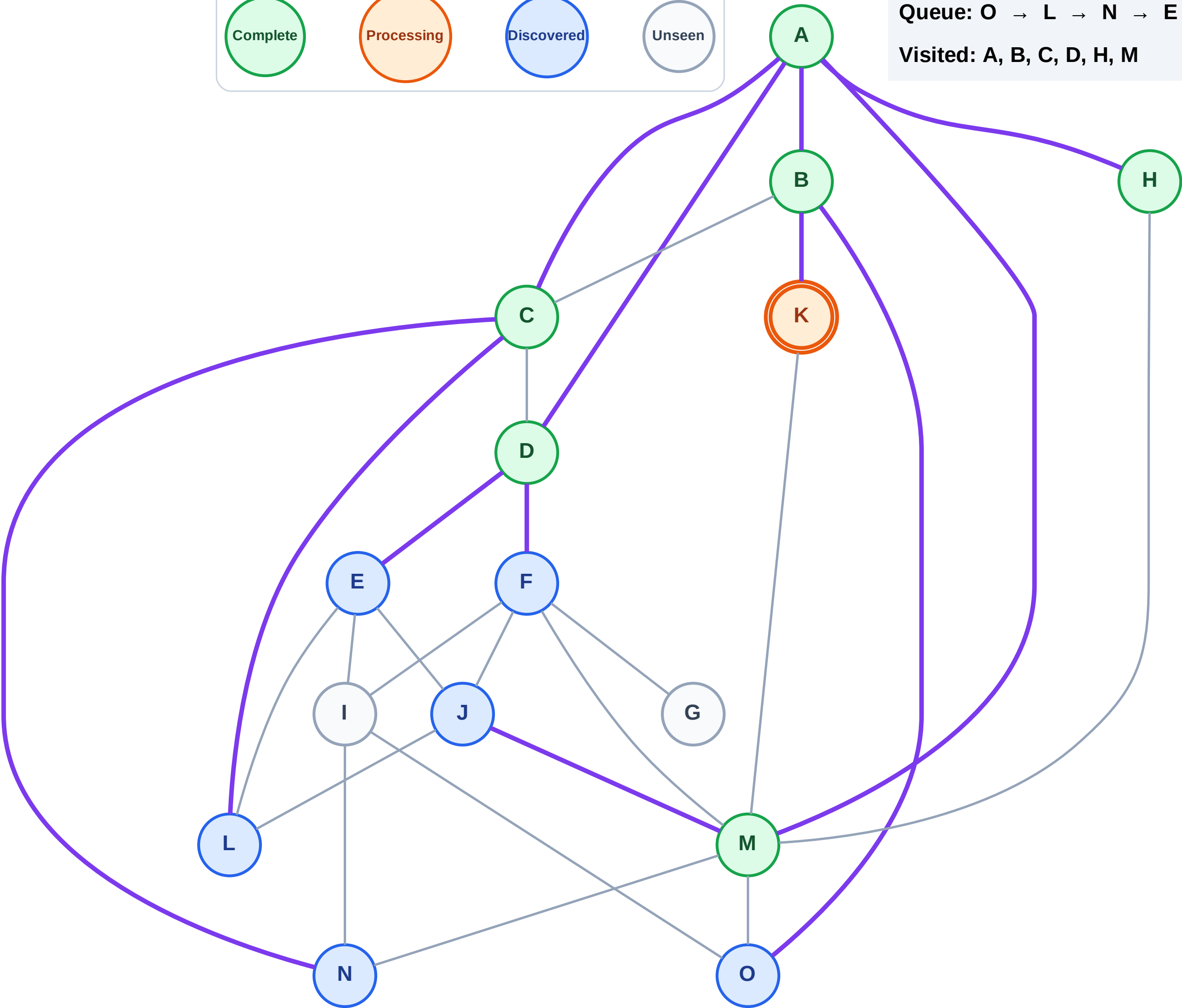
Processing

Discovered

Unseen

Queue: O → L → N → E → F → J

Visited: A, B, C, D, H, M



BFS Traversal

Step 15: TARGET NODE K REACHED! Path (2 hops): A → B → K

Legend

Complete

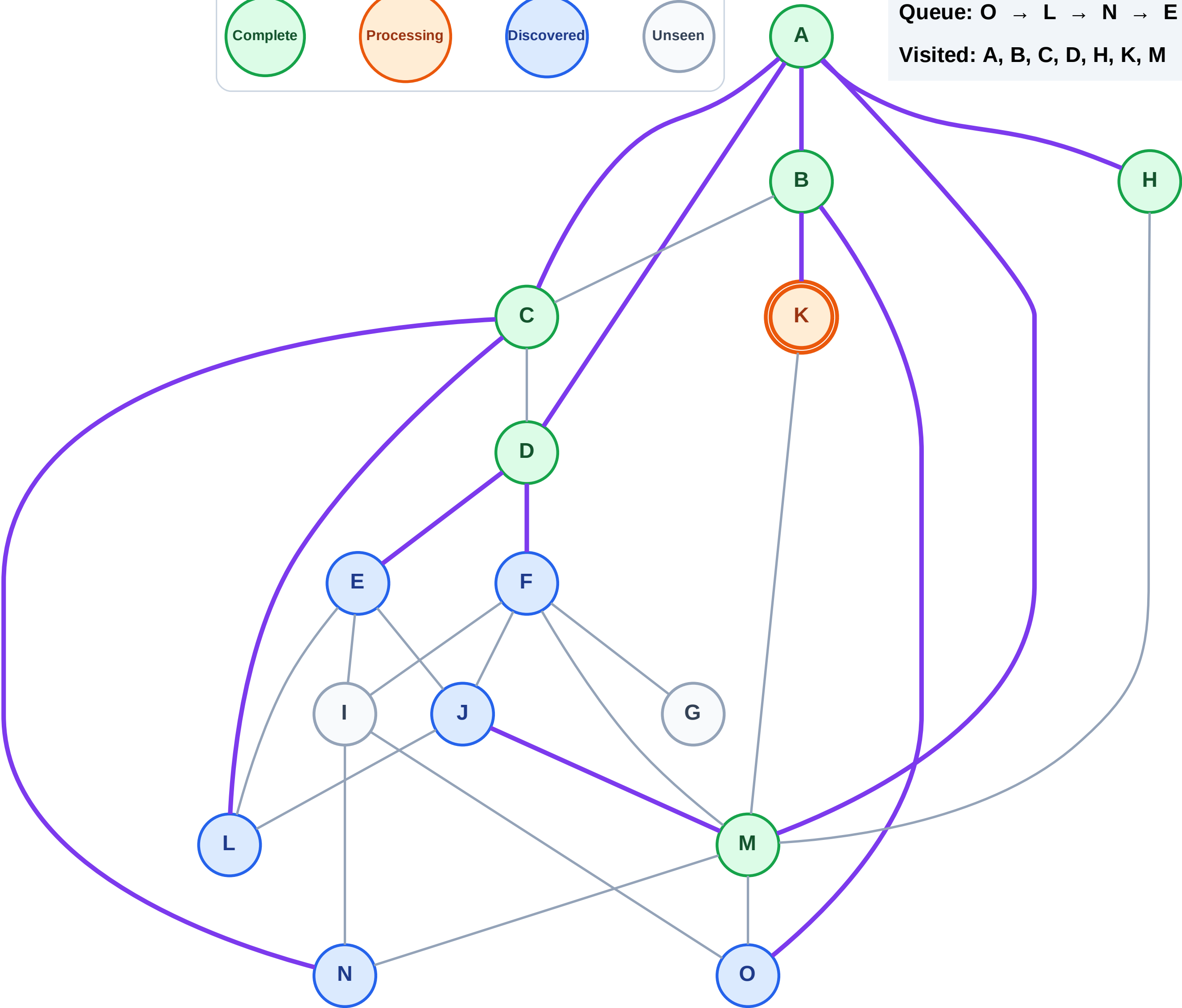
Processing

Discovered

Unseen

Queue: O → L → N → E → F → J

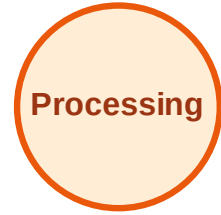
Visited: A, B, C, D, H, K, M



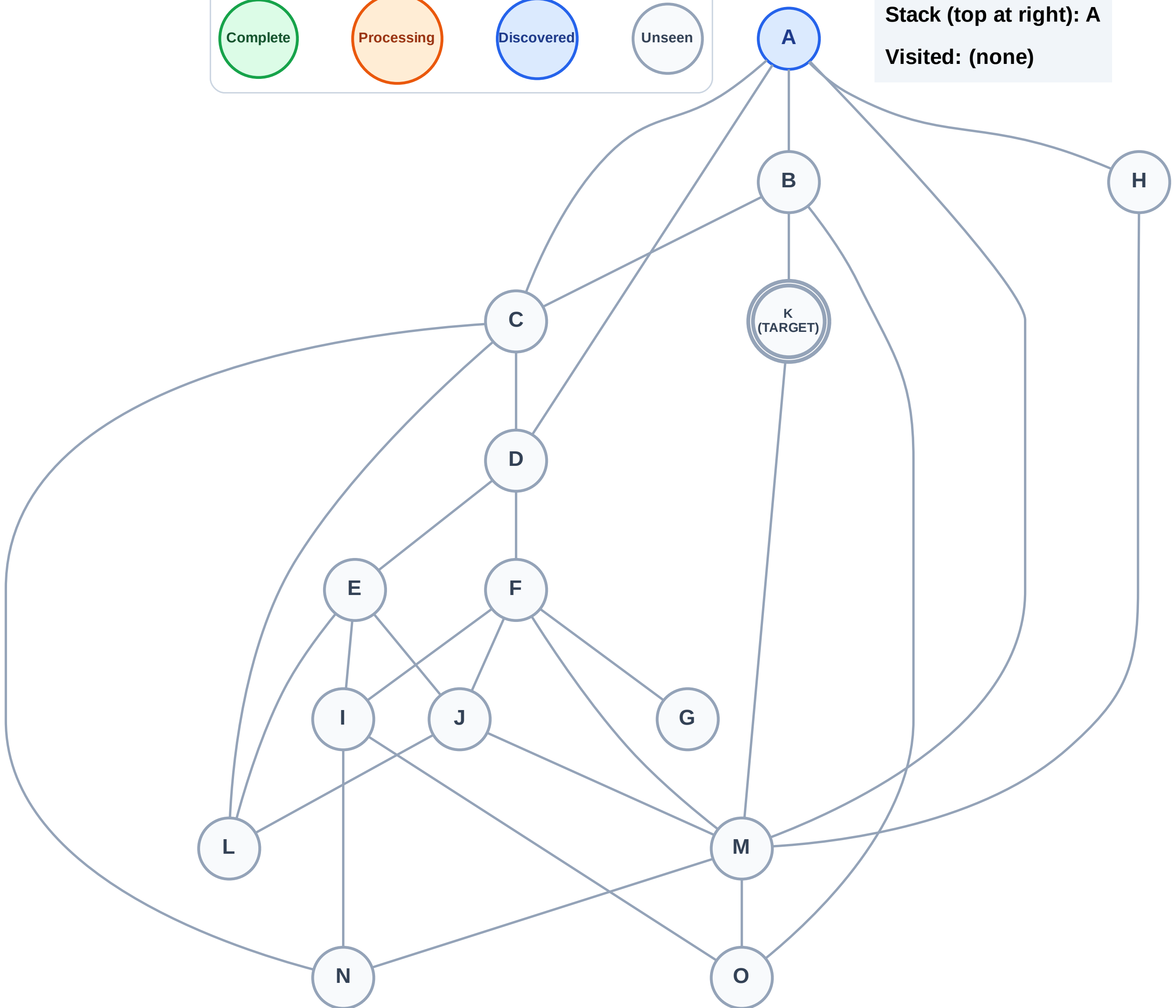
DFS Traversal

Step 1: Start at A (Seeking target node K)

Legend



Stack (top at right): A
Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

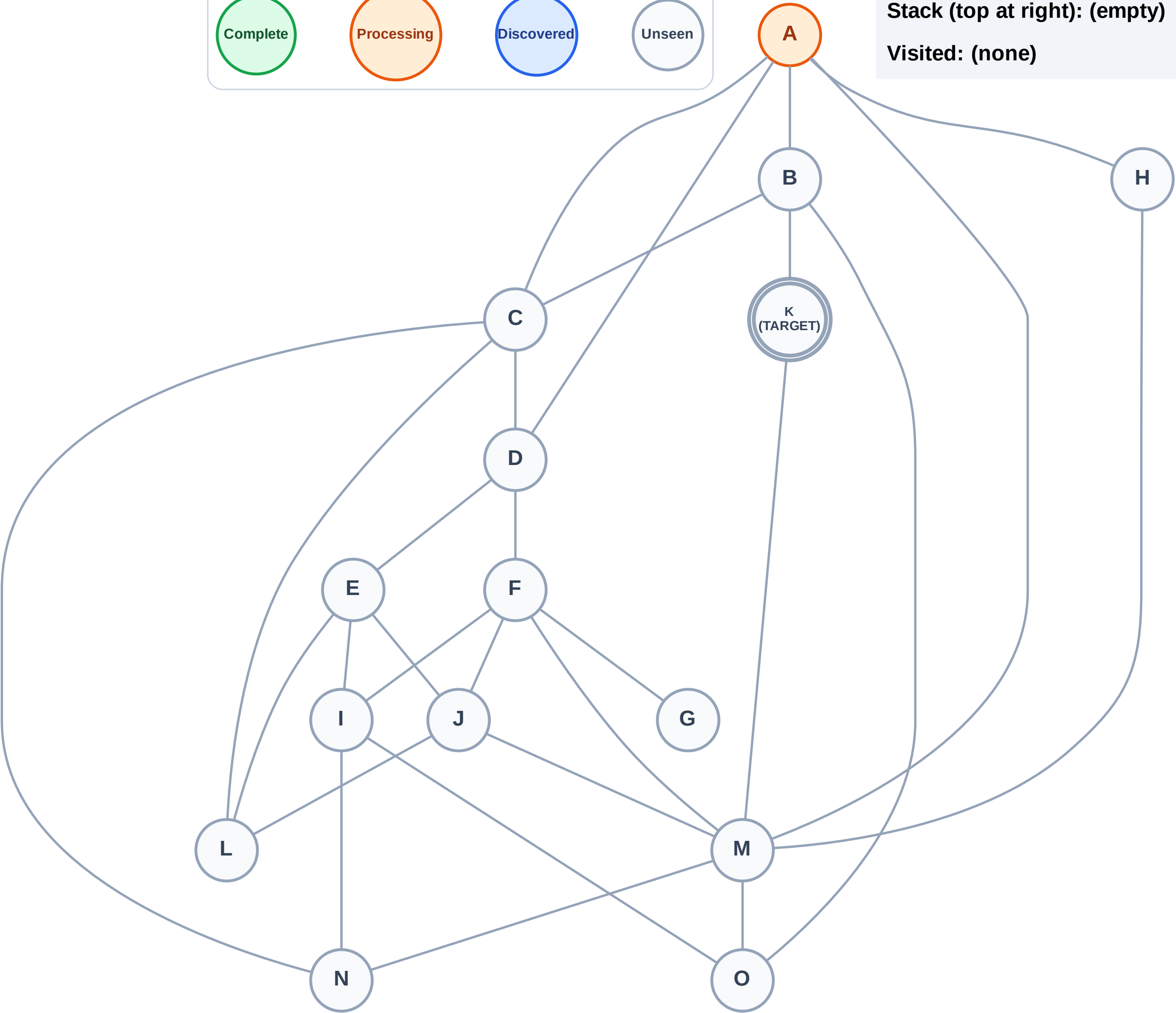
Complete

Processing

Discovered

Unseen

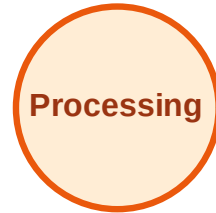
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

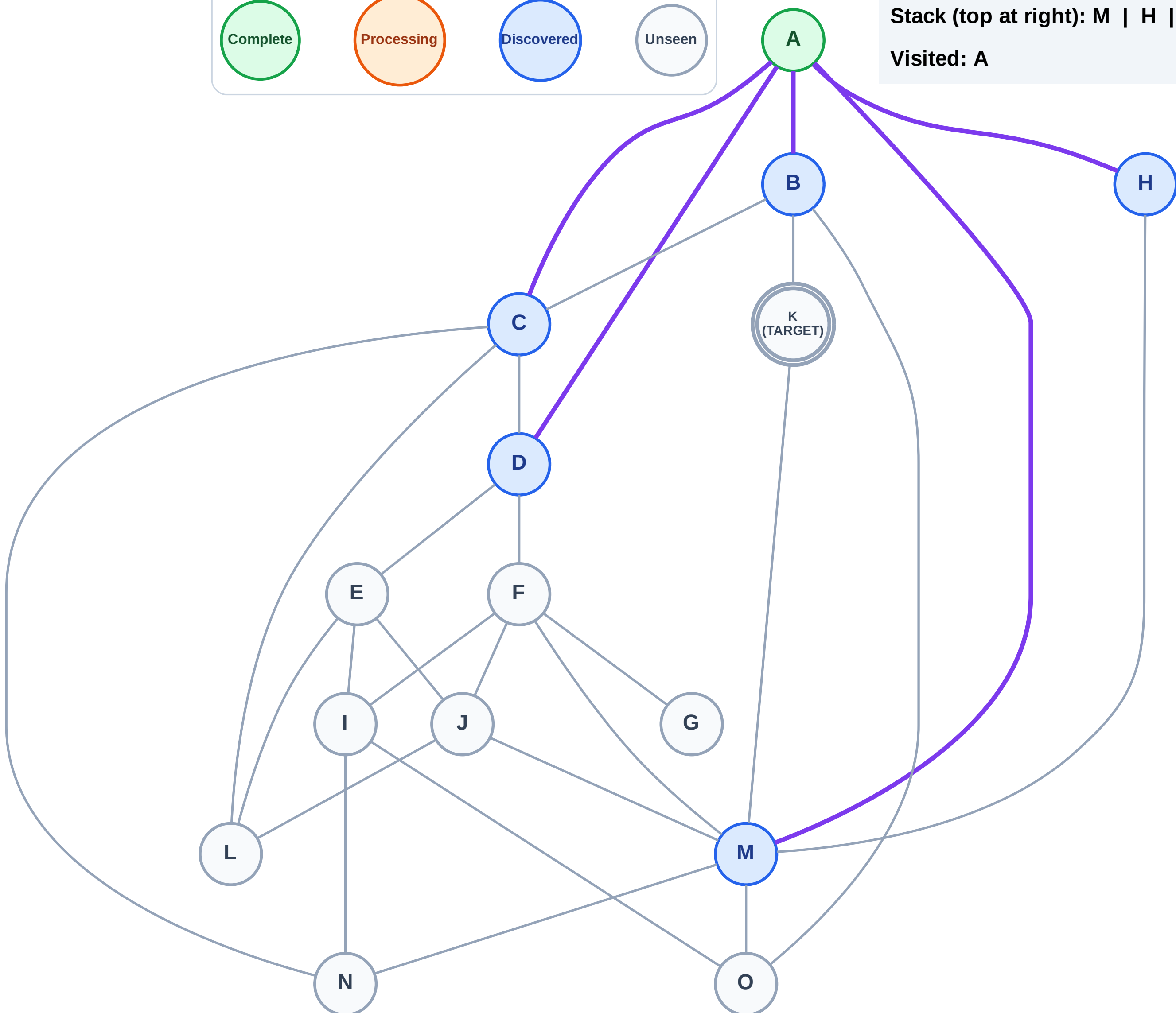
Step 3: Discover M, H, D, C, B from A

Legend



Stack (top at right): M | H | D | C | B

Visited: A



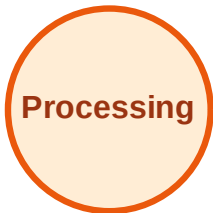
DFS Traversal

Step 4: Process node B

Legend



Complete



Processing



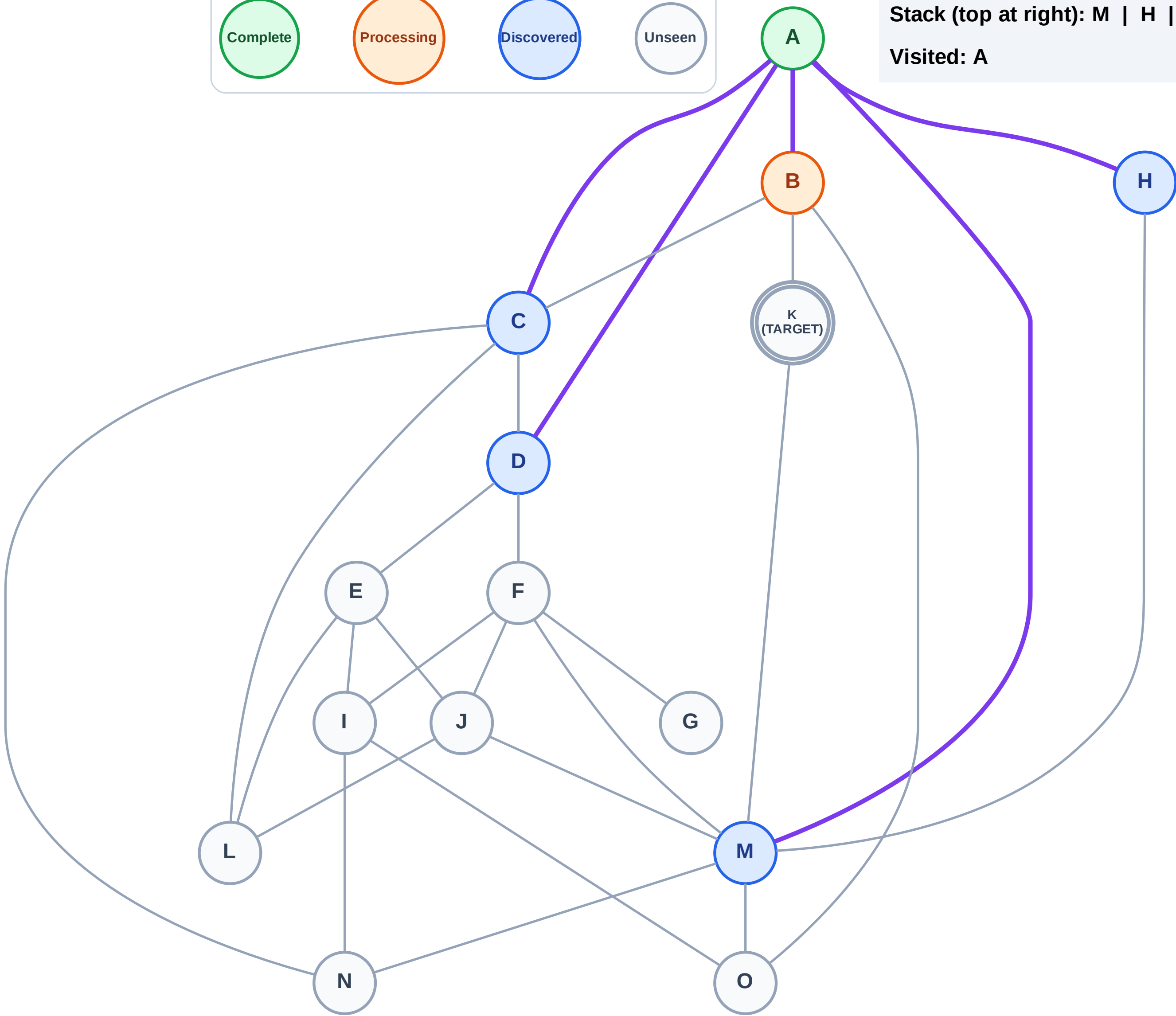
Discovered



Unseen

Stack (top at right): M | H | D | C

Visited: A



DFS Traversal

Step 5: Discover O, K from B

Legend

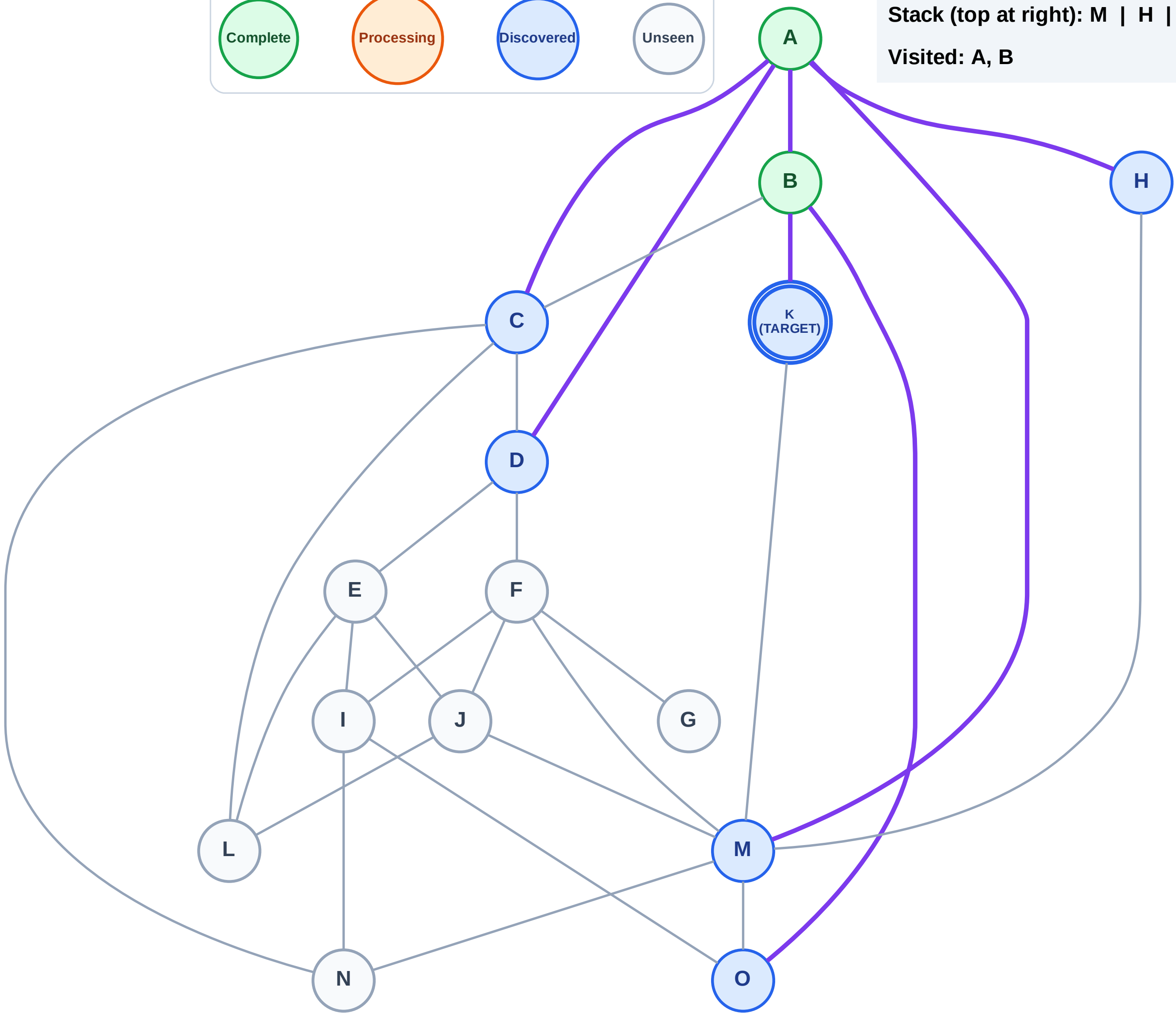
Complete

Processing

Discovered

Unseen

Stack (top at right): M | H | D | C | O | K
Visited: A, B



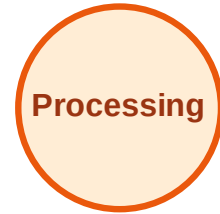
DFS Traversal

Step 6: Process node K

Legend



Complete



Processing



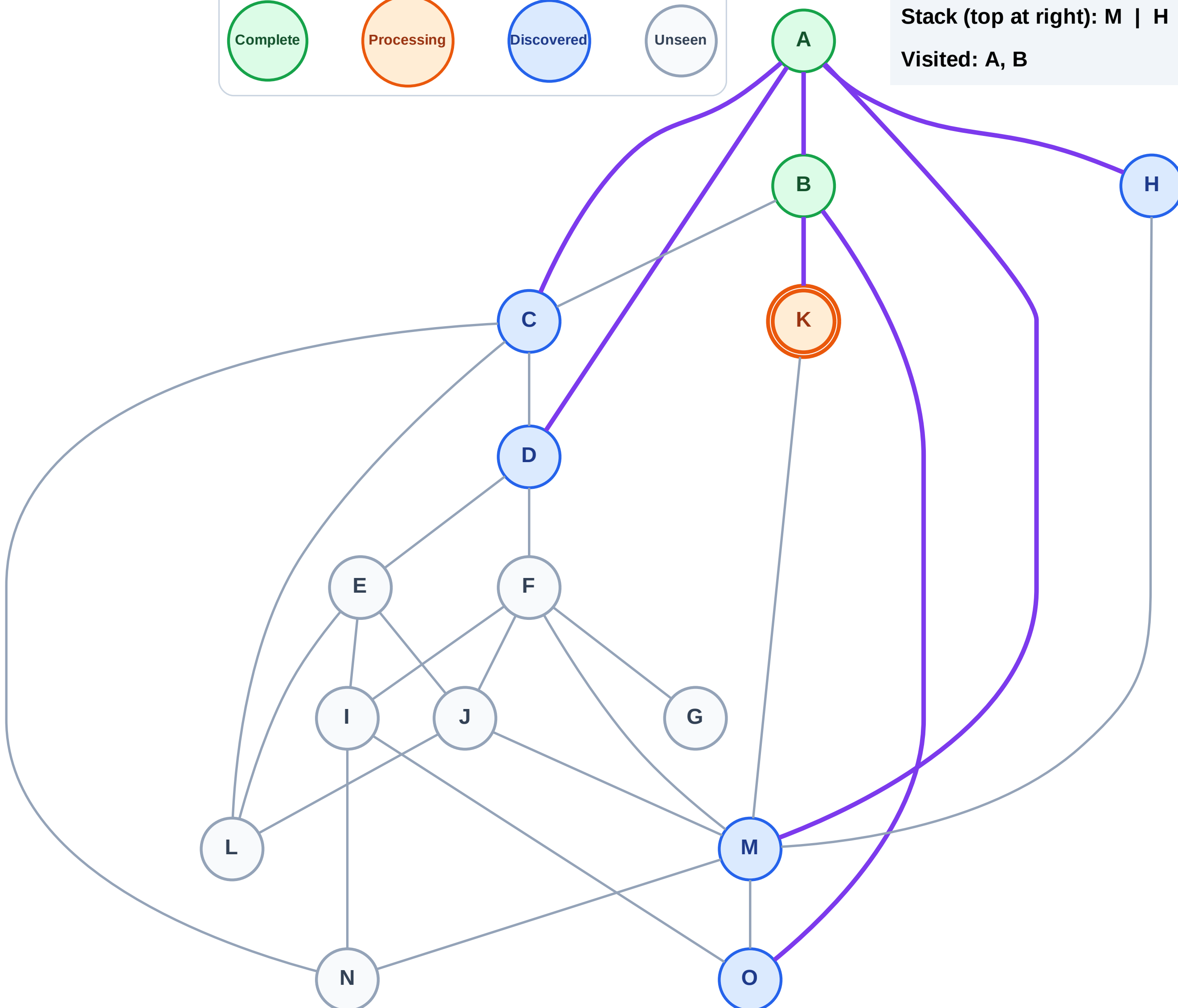
Discovered



Unseen

Stack (top at right): M | H | D | C | O

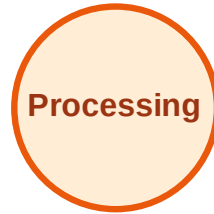
Visited: A, B



DFS Traversal

Step 7: TARGET NODE K REACHED! Path (2 hops): A → B → K

Legend



Stack (top at right): M | H | D | C | O

Visited: A, B, K

